PAUSEAI STUDY GUIDE: MODULE 1
NEURAL NETWORKS & FORWARD PROPAGATION

Overview:
A deep neural network consists of an input layer, hidden representations, and an output layer.
Each layer applies linear transformations parameterized by weight matrices and bias vectors.

Key Formula:
Linear activation $z = W*x + b$.
The forward pass maps input data through consecutive matrix multiplications to produce hypothesis predictions.

PAUSEAI STUDY GUIDE: MODULE 2
OPTIMIZATION, LOSS FUNCTIONS & GRADIENTS

Loss Landscapes:
The loss function quantifies model prediction discrepancies against ground-truth targets.
Stochastic Gradient Descent (SGD) introduces mini-batch stochasticity to escape local saddle points.
Learning rate scheduling is essential to avoid divergence or oscillating around steep ravines.

PAUSEAI STUDY GUIDE: MODULE 3
BACKPROPAGATION & AUTOMATIC DIFFERENTIATION

Mathematical Formulation:
Backpropagation calculates dL/dW through recursive chain rule expansion.
For layer l, error signal delta_l is computed from layer l+1.
Gradient clipping and momentum optimizers (Adam, RMSProp) stabilize deep backward passes.